The Drop Proof Rule: Six Gates and Four Responses

Macharia Barii

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Abstract

A single job is either provable at a price the customer will accept, or it is not. The Drop Proof Rule makes that test explicit as a sequence of six gates, evaluated cheapest-first: address, floor, contribution, density, schedulability, and risk. A job that passes all six is provable as presented; a job that fails a gate but sits within the customer's ceiling is priceable; a job whose floor exceeds the ceiling may be made provable by a change of state; a job for which no state change suffices is passed. Behind the gates sits the master inequality—the invariant $\bar{n}\,p(a)\,A(a) > c_{\text{mat}}+w_{\text{driver}}/R(a)+(1-p(a))c_r+\Phi C_p$ that recurs throughout the theory. This paper states the six gates precisely, derives the master inequality, enumerates the four responses, and gives the two complementary readings—the rule by drop type and the dual rule for the client book. The rule has exactly one proposition to remember: *"accept anyway" is not among the four responses.*

1 The Drop Proof Rule

The theory produces a single question for every job that the firm is asked to carry: is there a price at which this job contributes, and is that price one the customer will accept? The Drop Proof Rule is the answer to that question, stated as a decision procedure.

The rule is not a price. It is a proof. A job is *provable* if it passes six gates in sequence; the gates are ordered so that the cheap tests run first and the expensive ones run only when the job has survived the cheap ones. The six gates correspond to the six reasons a job can fail, and each gate failure names its own response.

2 Gate 1: Address

The first gate is the cheapest, because it requires only the place lookup from the scoring layer.

Definition 1 (Address gate)**.**

$$a \geq a_{\min}(n, T) \tag{1}$$

A job whose addressability a falls below the minimum for its net fee n and tier T cannot be served reliably at any price the customer will accept. Below $a_{\min}$ the first-attempt success is too low and the search time too high: the job cannot clear the master inequality no matter how it is priced. The correct response to a failing Gate 1 is **Prepare**—verify the address and raise the static component—not to cut the price, which cannot fix the search cost.

3 Gate 2: Floor

The second gate tests the rate card against its own floor.

Definition 2 (Floor gate)**.**

$$B \geq M \tag{2}$$

The base rate B produced by the rate card must clear the minimum drop charge M. If $B < M$, the customer is being quoted below the floor at which the drop covers its own dispatch. The response is **Price**: raise the base to the floor. This is the engine's refusal to dump a sub-floor price into the book.

4 Gate 3: Contribution

The third gate is the invariant, computed on the scoring layer.

Definition 3 (Contribution gate)**.**

$$\hat{p}\, A(a)\, B > c_{var} + (1 - \hat{p})c_r + \Phi C_p/N + \mathbb{E}[loss] \tag{3}$$

The expected net revenue of the drop—base rate B, discounted by first-attempt success $\hat{p}$ and lifted by the addressing surcharge $A(a)$—must exceed its expected cost: variable cost, reattempt cost, the per-drop share of the principal burden $\Phi C_p/N$, and expected cargo loss. This is the per-drop engine form of the master inequality (Section 8); in one equation it decides whether the drop should happen at all.

5 Gate 4: Density

The fourth gate asks whether the trip has enough drops to amortise its fixed cost.

Definition 4 (Density gate)**.**

$$N \geq n^* = \frac{c_0}{\hat{p}A(a)B - c_{var} - (1 - \hat{p})c_r - \Phi C_p/N} \tag{4}$$

The dispatch cost c_0 is fixed per trip, so the trip needs at least break-even density n^* to clear. A drop that passes Gates 1–3 in isolation can still fail here, because the fixed cost of the trip it rides on is spread over too few drops. The response is **Prepare**: consolidate the drop onto a denser trip in the same corridor.

6 Gate 5: Schedulability

The fifth gate tests whether a service window can be promised at all.

Definition 5 (Schedulability gate)**.**

$$\mathrm{Var}[\tau] \leq \Psi(W) \tag{5}$$

A window of width W is promisable only if the variance of service time is bounded by $\Psi(W)$. Below a certain addressability, no commercially useful window is promisable at any price—the product is infeasible, not merely unprofitable. The response is **Prepare** (accumulate delivery volume to shrink the variance) or, if no volume can be accumulated in time, **Pass**.

7 Gate 6: Risk

The sixth gate is the most expensive, because it requires the full loss model.

Definition 6 (Risk gate)**.**

$$\mathbb{E}[loss] < \mathit{contribution} \times \mathit{buffer} \tag{6}$$

Expected cargo loss must be bounded by a buffer multiple of the drop's contribution. For high-value cargo the loss term can exceed the entire delivery fee, so this gate—not the contribution gate—binds. The response is **Prepare**: add security tier, or route through a verified recipient with an established trust record.

8 The Master Inequality

The six gates are the operational form of a single invariant that appears throughout the series:

$$\boxed{\bar{n}\, p(a)\, A(a) > c_{\text{mat}} + \frac{w_{\text{driver}}}{R(a)} + (1 - p(a))c_r + \Phi C_p} \tag{7}$$

Read plainly: the average net fee, discounted by first-attempt success and lifted by the addressing surcharge, must exceed materials cost, plus labour cost per drop (which is the driver wage w_{driver} divided by the drop rate $R(a)$), plus expected reattempt cost, plus the principal burden per trip.

The two forms are the same inequality at different resolutions. Gate 3 writes the per-drop terms as $c_{\text{var}} + \Phi C_p/N + (1 - \hat{p})c_r$; the master inequality writes them as $c_{\text{mat}} + w_{\text{driver}}/R(a) + (1 - p(a))c_r + \Phi C_p$. The labour term $w_{\text{driver}}/R(a)$ is distributed in the engine across variable cost and the density term; the principal burden ΦC_p is the per-trip quantity whose per-drop share falls as density rises. Everything downstream—the floor, the fixed point, the gates—is a rearrangement of this one inequality.

The gates are tested cheapest-first, so that a job that will fail a late gate is rejected before the expensive computation. Gate 1 requires only the place lookup; Gate 6 requires the full loss model.

9 The Four Responses

The verdict is one of four, and the choice between them is where discipline lives.

Response	Condition	Meaning	Correct action
Prove	All gates pass	Job is provable as presented	Accept at $\max(p^*, B)$
Price	Floor $\leq$ ceiling; some gate fails	Interval exists; the gate failure is overcome by price	Accept at p^*
Prepare	Floor $>$ ceiling, but a state change closes the gap	The job is in the wrong state	Verify address, consolidate, add security, or change client mix
Pass	No state change closes the gap	The job is not provable at any acceptable price	Decline

"Accept anyway" is not among the four. Every job that enters the book without passing its gates is a subsidy from the profitable jobs to the unprofitable ones. That subsidy is what killed the previous generation of logistics companies, who could not produce the verdict the engine is built to produce: *no stable price exists.*

10 The Rule by Drop Type

The same six gates resolve differently for different drop types, and the differences are what make the rule a segmentation tool rather than a blunt threshold.

Mass-market drops. Low net fee ($\bar{n} \approx 800$) and low addressability together place the mass market below the static threshold $a^* \approx 0.36$. Informal urban micro-retail sites sit at $a \approx 0.2$–0.3. Gate 3—and therefore the master inequality—fails regardless of volume. The correct verdict is **Pass**, unless the address is first verified (Gate 1) to lift the static component above threshold.

B2B contract drops. A 2.5× increase in drop value ($\bar{n} = 2{,}000$) lowers the threshold to $a^* \approx 0.17$, below the informal baseline. Gate 3 clears at low addressability; the book is provable

where the mass market is not. The verdict is **Prove**, and the high-value B2B segment becomes the data-acquisition strategy that funds the ledger.

High-value drops. Where declared value is high, expected loss becomes the dominant term: a 3% loss rate on a KSh 50,000 device is KSh 1,500 per drop, larger than the entire delivery fee. Gate 6—not Gate 3—binds, and its response is not a price but a security tier and a verified recipient. The rule channels high-value volume through the secured tier and the recipient trust graph, where expected loss is governed by recipient history rather than by place addressability.

11 The Dual Rule (client book)

The six gates test a single drop. The dual rule tests the client whose book the drop belongs to, and it is the mirror image of the same inequality.

A client's book is provable as a whole if and only if the sum of expected contributions across its drops clears the sum of expected costs:

$$\sum_{j \in \mathcal{C}} \bar{n}_j \, p(a_j) \, A(a_j) \;>\; \sum_{j \in \mathcal{C}} \left[c_{\mathrm{mat},j} + \frac{w_{\mathrm{driver}}}{R(a_j)} + (1 - p(a_j)) c_r \right] + \Phi C_p \cdot |\mathcal{C}| + \sum_{j \in \mathcal{C}} \mathbb{E}[\mathrm{loss}_j] \qquad (8)$$

The dual rule is equivalent to a single book-level test on the volume-weighted mean addressability $\bar{a}$ of the client's placements: the book clears exactly when its weighted-average drop sits above the threshold. The practical form is the client-selection criterion that runs through the theory: qualify clients on *expected drops per address*, not merely on revenue. A manufacturer shipping daily to twelve depots is deep and compounding; an e-commerce client shipping once to five thousand homes is broad, shallow, and decaying. The dual rule refuses or prices up the third category early—the pattern, in the theory's terms, of high coverage and zero depth.

Taken together, the six gates answer "shall this drop proceed," and the dual rule answers "shall this client remain on the book." They are the same inequality examined at two distances.